

Weather Wise AI

Phase 2: Requirement Analysis

This phase defines exactly what Weather Wise AI must do (functional) and how well it must do it (non-functional), along with the resources needed.

Functional Requirements

- User can search any city or use GPS to get current weather.
- Show hourly (48 h) and daily (7 day) forecasts with temperature, rain chance, wind, humidity, UV and AQI.
- AI chat: user asks natural-language weather questions and receives clear answers.
- Smart alerts and recommendations based on user preferences.
- Save favourite locations and set alert preferences (user profile).

Non-Functional Requirements

- Performance: results load in under 3 seconds.
- Accuracy: forecast error should improve over a plain API baseline.
- Availability: 99% uptime; graceful fallback if the data API fails.
- Security and privacy: location data used only with consent; HTTPS everywhere.
- Usability: responsive design for mobile and desktop.

Data and Technical Requirements

Category	Requirement
Data sources	Weather API (e.g. OpenWeatherMap or Open-Meteo), historical weather datasets, air quality API
AI / ML	Python, scikit-learn / TensorFlow for forecasting; LLM API for the conversational assistant
Backend	Python (Flask or FastAPI), REST APIs
Frontend	React or HTML/CSS/JS, chart library for graphs
Database	SQLite/PostgreSQL for users and saved locations
Tools	Git/GitHub, VS Code, Postman, Jupyter Notebook

Constraints and Assumptions

- Free API tiers have request limits, so caching is required.
- Users have internet access; offline mode shows last cached data.